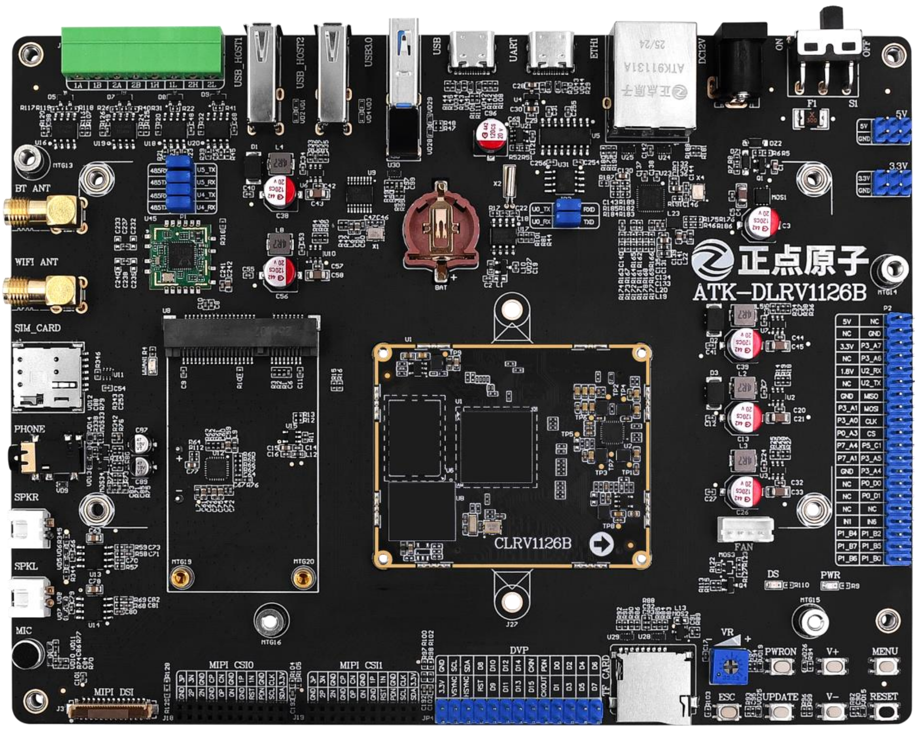
ATK-DLRV1126B

Development Board Specification

V1.0



ALIENTEK

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Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2025.12

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Chapter 1. Overview of Development Board

1.1 Introduction to the ATK-DLRV1126B Development Board

The ALIENTEK RV1126B development board is based on the domestic Rockchip RV1126B/BJ processor, equipped with a quad-core high-performance chip: 4×Cortex-A53 (up to 1.6 GHz), and an NPU with 3 TOPS computing power. It supports a 2D Graphics Engine (RGA), 4K hardware codec, and ISP.

It features powerful edge AI computing capabilities and supports INT8/INT16 mixed-precision computing. It covers core vision task networks such as object detection, semantic segmentation, and image classification. Meanwhile, it integrates advanced voice processing capabilities, including highly natural text-to-speech (TTS) and accurate automatic speech recognition (ASR) technologies, and provides a full-process OCR text detection and recognition solution. The system is deeply compatible with mainstream deep learning frameworks including PyTorch, TensorFlow, and ONNX.

Supports H.265/H.264 hardware decoding up to 4K (3840x2160) @30fps. Supports HEVC (H.265) Main Profile Level 5.0 High Tier, H.264 High Profile Level 5.0, and JPEG Baseline up to 12M@30fps. Supports hardware encoding for mainstream resolutions including 4K (3840x2160) @30fps and 1080P@30fps. It can be applied in smart security, smart door locks, industrial vision, smart home, intelligent automotive, robotics, and other fields.

The development board adopts a separate core board + baseboard design. The core board connects to the baseboard via a BTB connector. The baseboard integrates 2× USB2.0 HOST, USB3.0 HOST, USB OTG, Type-C debug serial port, Gigabit Ethernet port, external RTC, TF card slot, 2× MIPI CSI, MIPI DSI, microphone, speaker and earphone interfaces, 4G module interface, WiFi & Bluetooth module, 2× RS485, 2× CAN FD. It features rich interface functions.

The ATK-DLRV1126B development board is supplied with abundant Linux system development documents and open software resources. Enterprise customers can directly purchase the core board for their own product R&D, and ALIENTEK provides a complete and comprehensive system source code to facilitate the product R&D of enterprise customers. To improve the development efficiency of enterprise users and shorten the development cycle, ALIENTEK has specially compiled a series of materials for core board users that are required during the development phase, covering schematic diagrams, baseboard design data, mechanical structures, component packages, connector specifications, factory system image source code, compilers, software packages and more, for the convenience of enterprise users' development.

Resource Download:

Download Center: http://www.openedv.com/docs/boards/arm-linux/index.html

1.2 Application Areas

It is equipped with the RV1126B, a 4K universal visual AI vision processor, with an NPU computing power of up to 3.0 TOPS. It supports hardware decoding of H.265/H.264 at a maximum of 4K (3840x2160) @30fps, as well as hardware encoding for mainstream resolutions including 4K (3840x2160) @30fps and 1080P@30fps. It can efficiently implement detection and recognition, and is applicable to lightweight AI scenarios such as license plate recognition, face detection, face tracking, and image classification.

Application Scope of RV1126B:

- Industrial Vision and AIoT Edge Computing: Integrating an NPU and a high-performance image processing unit, it supports machine vision applications such as face recognition, behavior analysis and defect detection. It is suitable for scenarios like industrial quality inspection, smart security and equipment monitoring, enabling real-time intelligent decision-making at the edge.
- Smart Display and Interactive Terminals: Supporting multi-channel camera input and high-definition display output, it features H.264/H.265 hardware encoding and decoding capabilities, can drive MIPI screens and run the QT graphics framework smoothly. It is applicable to high-definition human-machine interfaces such as smart retail, interactive terminals and industrial HMIs.
- Multimedia Information Processing and Transmission: Supports multi-channel video capture and synchronous processing, with intelligent encoding and network streaming capabilities. It can realize applications such as remote monitoring, video conferencing, and live streaming, meeting the demands of real-time audio and video processing and low-latency transmission in industrial and consumer scenarios.
- Industrial Control and Communication Gateway: Equipped with rich peripheral interfaces (Ethernet, UART, CAN, USB, etc.), it can build an integrated industrial gateway for data collection, protocol conversion and edge computing, facilitating device networking and data uploading to the cloud.

Chapter 2. Development Board Hardware Parameters

2.1 ATK-DLRV1126B Development Board

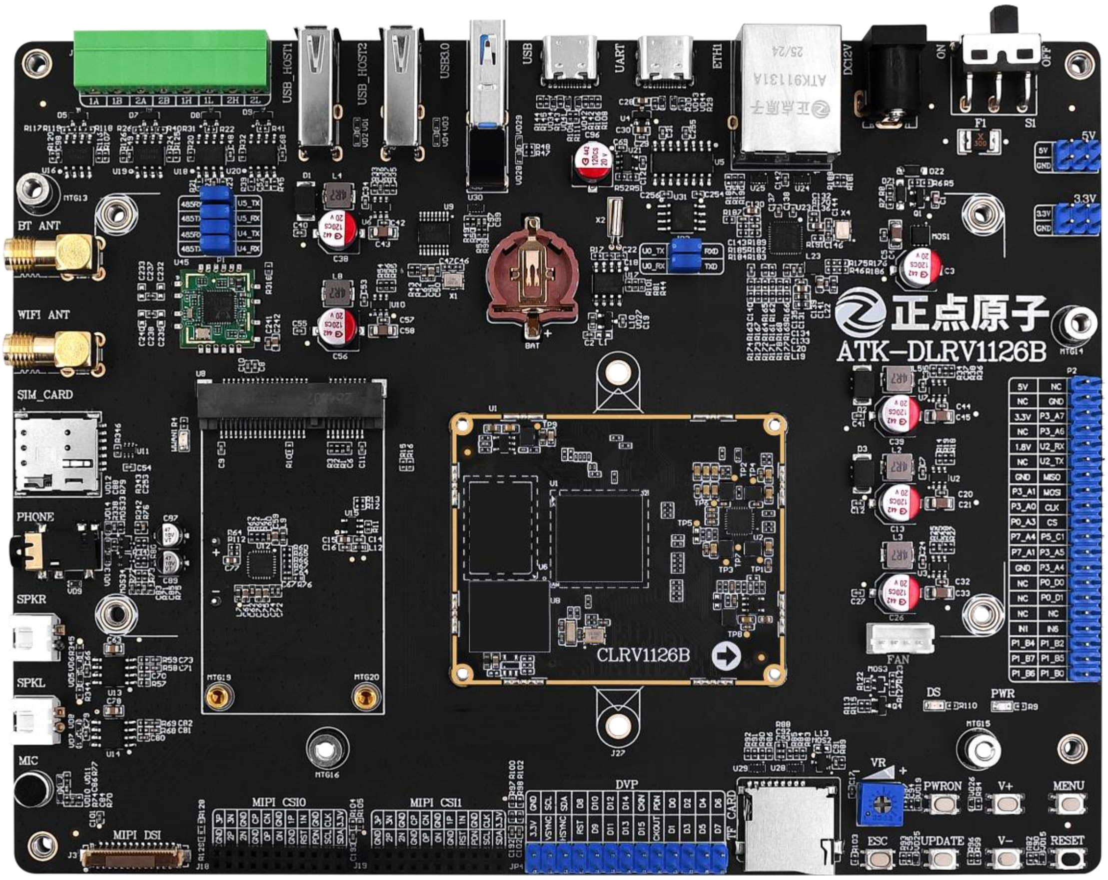

Figure 2.1-1 Front view of the development board

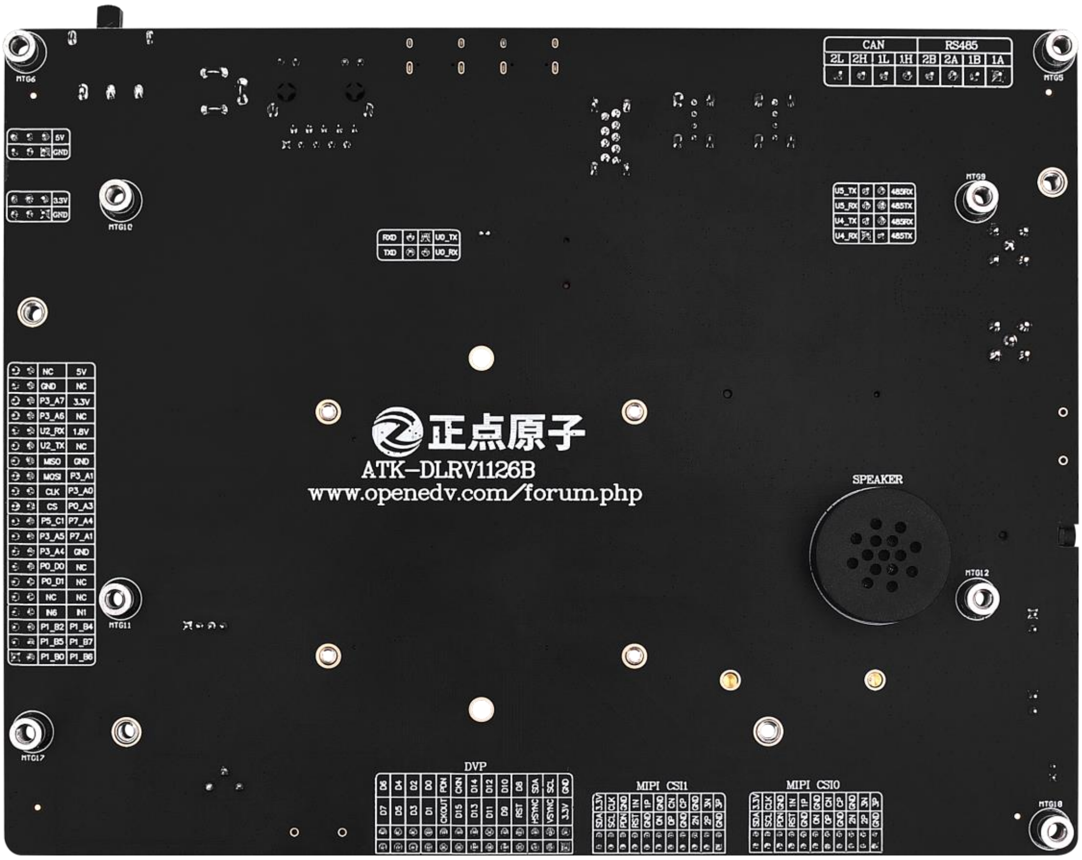

Figure 2.1-2 Back side of the development board

2.2 Hardware Parameters of the ATK-CLRV1126B Core Board

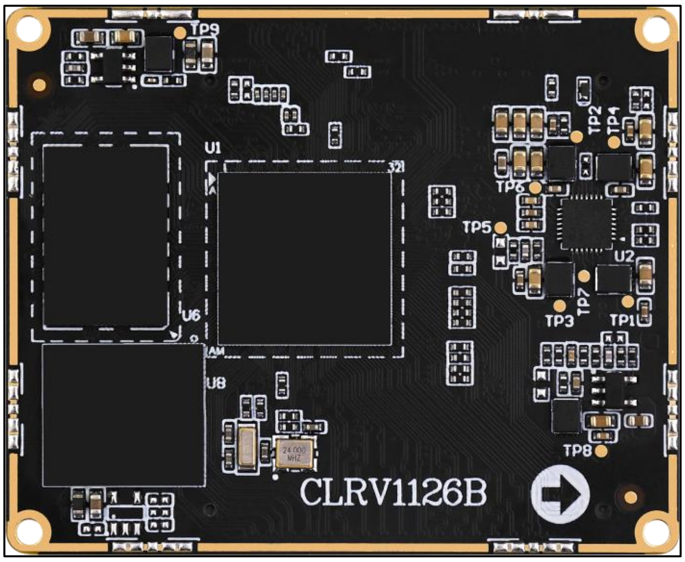

Figure 2.2-1 Front view of the core board

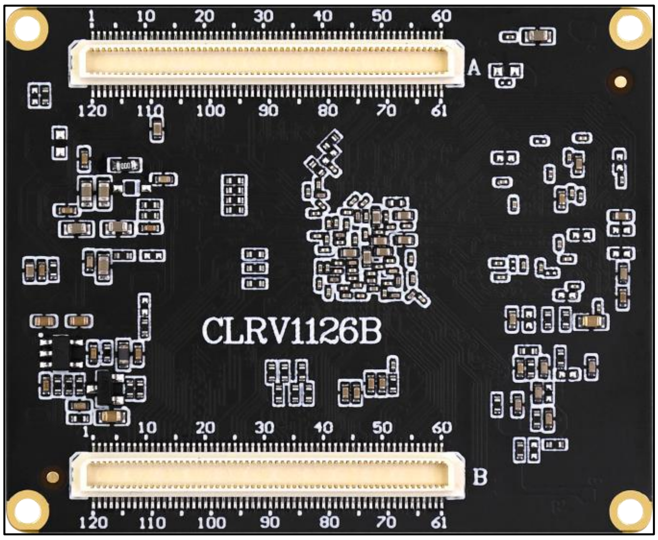

Figure 2.2-2 Back side of the core board

Tiem	Parameter	Remarks
Dimensions	55mm*45.00mm	Length × Width
CPU	RV1126B	BGA491 package
Memory	1GB/2GB DDR	On-board package. Due to chip supply, different manufacturers' chips may be used; refer to the actual mounted model.
Storage	8GB/32GB EMMC	On-board package. Due to chip supply, different manufacturers' chips may be used; refer to the actual mounted model.
Operating Voltage	5.0V	
Power Consumption	>1.8W	Reference value; actual power consumption depends on peripherals.
Operating Temperature	Commercial core board: 0 °C ~ +70 °C Industrial core board: -40 °C ~ +85 °C	RV1126B is commercial grade, RV1126BJ is industrial grade; the two are compatible.
Pin Count	240pin	
Pin Pitch	0.5mm	
Core Board Connection	BTB (Board-to-Board)	
PCB Technology	8-layer, gold-plated, independent ground and signal layers	Lead-free process

2.3 Hardware Parameters of the ATK-DLRV1126B Development Board Baseboard

Item	Parameter	Remarks
Dimensions	180mm*140mm	Length × Width
PCB Technology	4-layer, independent ground and signal layers	
Operating Voltage (1)	12V	
Power Consumption(2)	≥1.8W	Static power consumption; actual power consumption depends on peripherals.

2.4 On-board Resource Parameters of the ATK-DLRV1126B Development Board

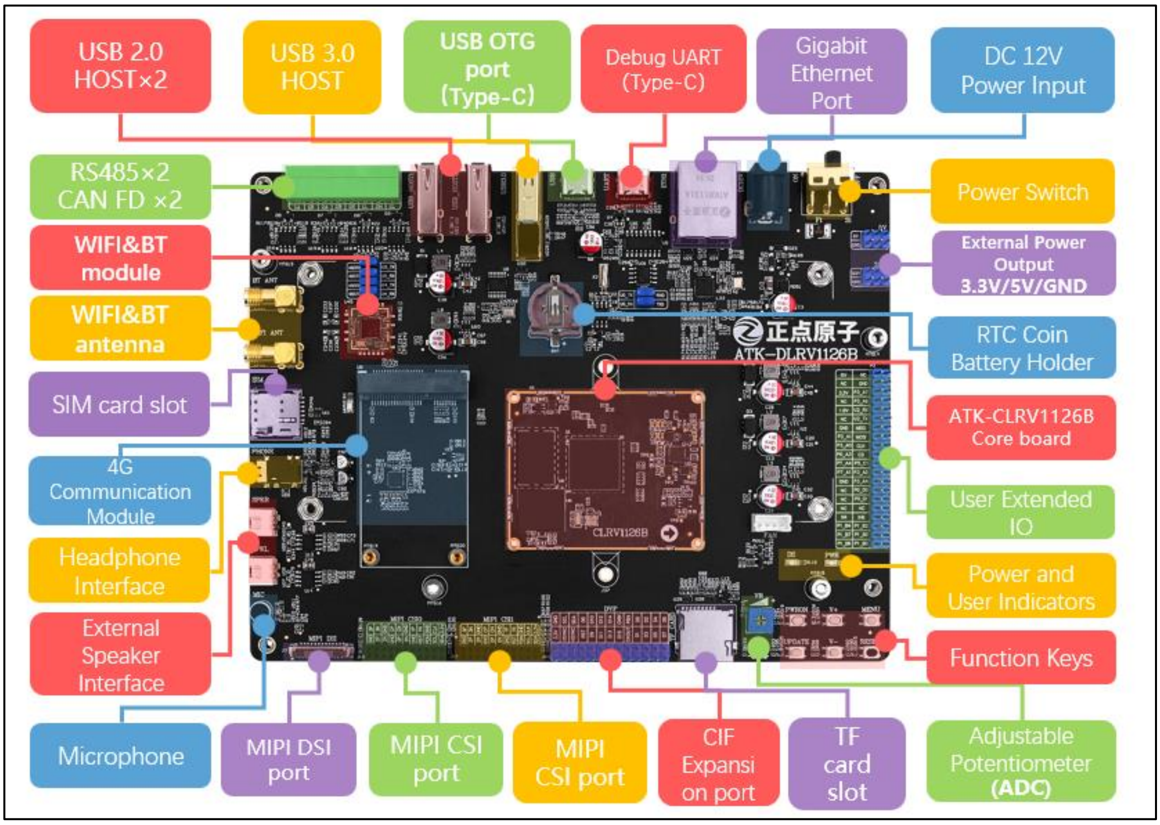

Figure 2.4-1 On-board Resource Diagram of the Development Board

- 1 core board connector
- 1 DC power input port (DC 12V)
- 1 MIPI DSI channel: supports 720×1280, 800×1280, 1080×1920 MIPI screens sold in the store
- 2 MIPI CSI channels: supports IMX415, IMX335 cameras sold in the store
- 1 Gigabit Ethernet port: 10/100/1000M RGMII
- 1 WiFi & Bluetooth module
- 1 4G module interface: supports EC20 module sold in the store
- 3 UART channels: 1 Type-C debug serial port, 2 RS485 ports
- 2 CAN channels: supports CAN FD
- USB interfaces: USB3.0 HOST (shared with OTG programming port), 1 USB2.0 HSOT (2 USB ports)

- 1 adjustable potentiometer
- 1 TF card slot
- 1 external RTC
- 1 3.5mm headphone jack
- 1 microphone
- 3 function keys, 4 programmable keys
- 1 programmable LED
- 1 speaker
- 1 fan connector
- 1 40-pin header (refer to the baseboard schematic for detailed pin functions)

2.5 RV1126B Chip Parameters

The summary table of partial resources of the RV1126B main control chip is as follows:

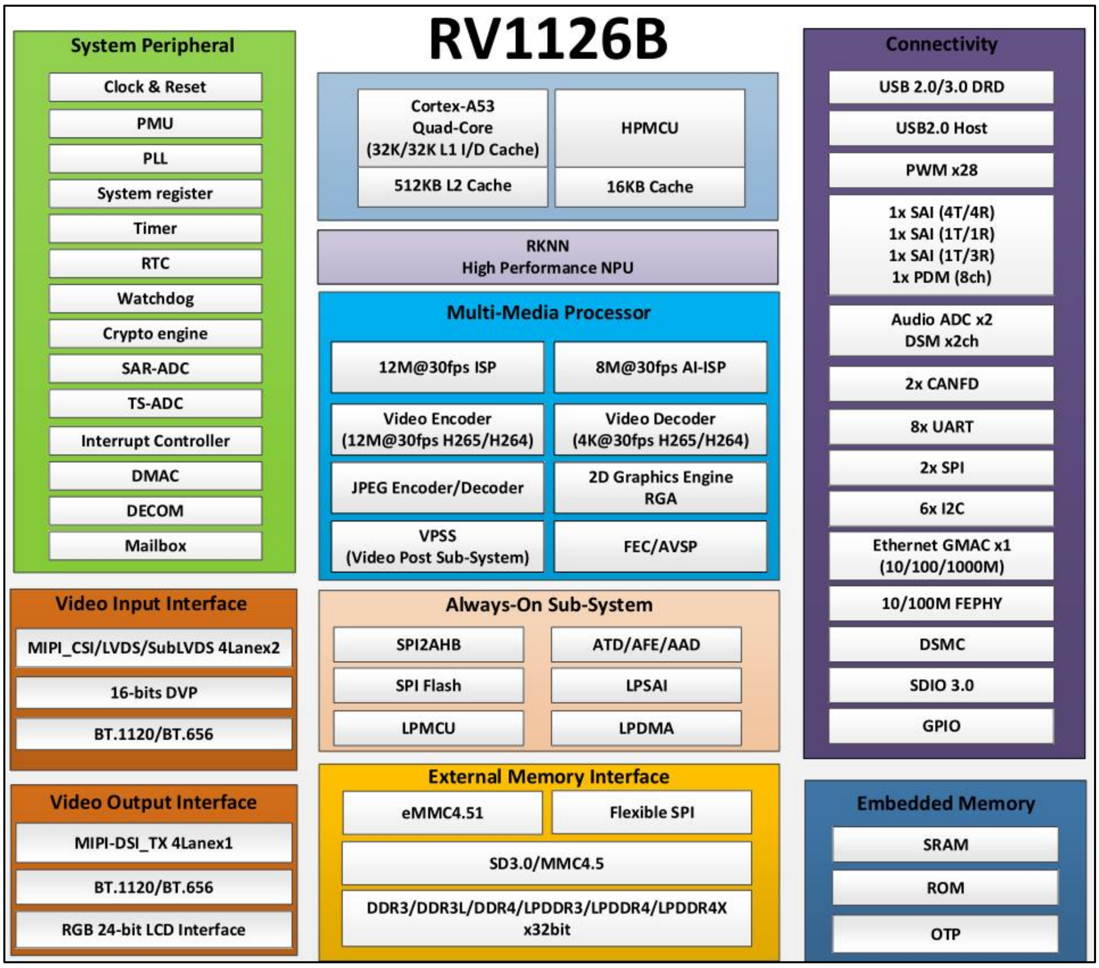

Figure 2.5-1 Chip Resources

For detailed specifications, please refer to the chip datasheet provided by Rockchip.

RV1126B Main Controller Chip Resources	
Processor	Quad-core ARM Cortex-A53

NPU	3.0 TOPs (supports INT8/INT16 mixed-precision computing)
ISP	Supports HDR, 3A, LSC, 3DNR, 2DNR Max input: 12M@30fps
RGA	Input formats: ARGB8888, RGBA8888, RGB888, RGB565, YUV422/420/444 (Planar/Semi-planar, 8/10-bit) and other mainstream formats. Output formats: ARGB8888, RGBA8888, RGB888, RGB565, YUV420/422/444 (Planar/Semi-planar), YUV400. Max resolution: 8192×8192 source, 4096×4096 destination
Video Decoder	H.264、H.265：3840x2160@30fps
Video Encoder	H.265, H.264: Supports hardware encoding at mainstream resolutions including 4K (3840×2160)@30fps and 1080P@30fps
JPEG Encoder	JPEG Baseline encoding Supports simultaneous encoding (e.g., HEVC+JPEG or H.264+JPEG) Max support: 12M@30fps
JPEG Decoder	Supports YUV400/YUV420/YUV422/YUV440/YUV411/YUV444 Supported image size range: 48×48 to 65520×65520
Display Output Interfaces	MCU/RGB LCD interface: up to 24 bits BT.656/BT.1120 interface 4-lane MIPI interface: 1.5 Gbps/lane, supports up to 1920×1080@60fps
Video Input Interfaces	2× MIPI CSI, each with MIPI DPHY V1.2, 2 lanes, up to 2.5 Gbps per lane
Audio Interfaces	3× SAI: SAI0 supports 4TX and 4RX, SAI1 supports 1TX and 1RX, SAI2 supports 1TX and 3RX
Ethernet Interface	1× Ethernet port: Integrated Ethernet PHY supporting 10/100 Mbps compliant with IEEE 802.3/802.3u. Supports 10/100/1000 Mbps via RGMII interface.
USB Controller	1× USB2.0 Host: supports High-Speed (480 Mbps), Full-Speed (12 Mbps), and Low-Speed (1.5 Mbps) modes. 1× USB3.0
SPI	2× SPI
I2C	6× I2C, supports 7-bit and 10-bit addressing; Data rates: Standard mode up to 100 kbps, Fast mode up to 400 kbps, High-Speed mode up to 1 Mbps.
UART	8× UART
PWM	4× PWM
CAN	2× CAN, supports CAN and CAN FD
ADC	3× SARADC (13-bit resolution, max sampling rate 2 MS/s)

Note: The values listed here are the resource parameters from the chip datasheet, not the available resource parameters of the core board.

Chapter 3. Product Specifications

3.1 Mechanical Dimensions of the Development Board

The development board has an overall dimension of 180 mm × 140 mm. The design is user-oriented and finalized through multiple improvements, based on Punctual Atom’s years of experience in development board design.

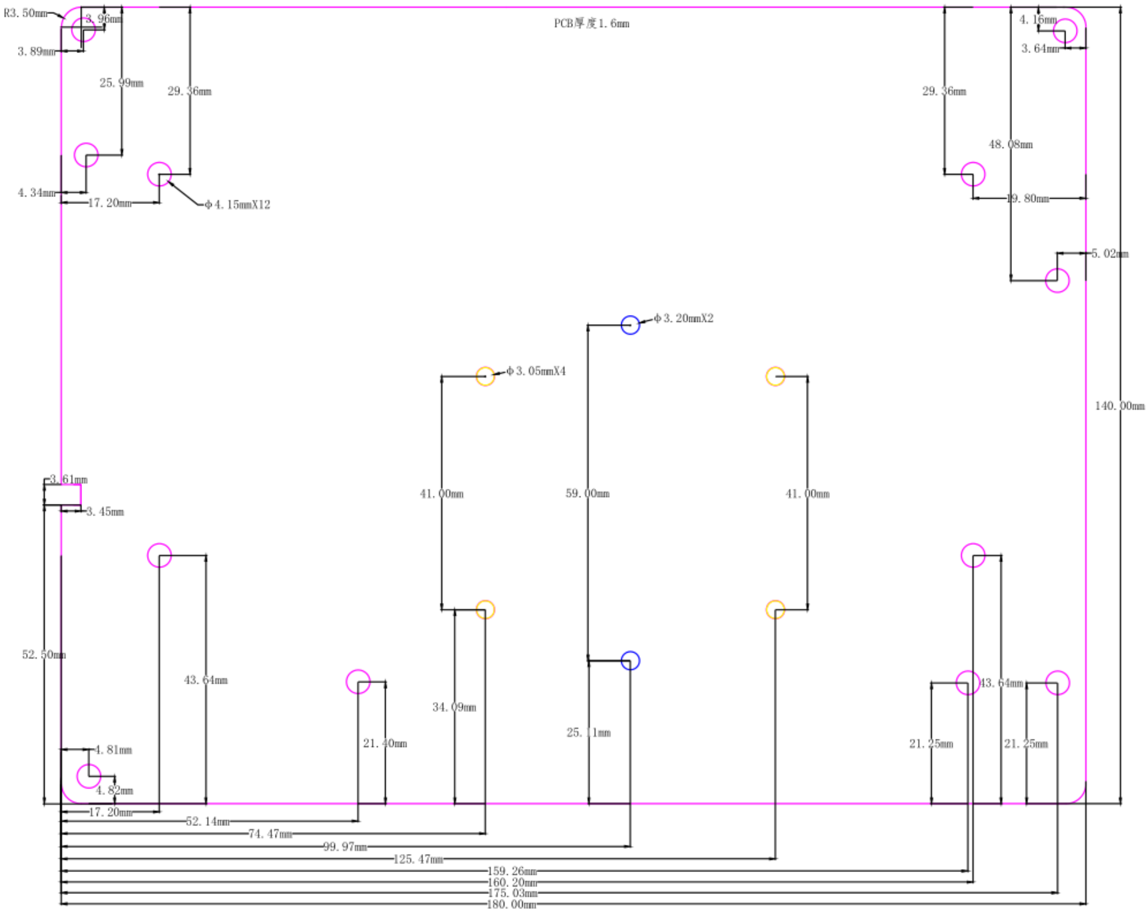

Figure 3.1-1 Mechanical dimensions of the development board

3.2 Software resources of the development board

The factory-default Linux system software resources are listed in Table 3.2.1 below:

Type	Description	Remarks
U-Boot	Version: 2017.09	Provided in the resource package
Linux Kernel	Version: 6.1	Provided in the resource package
Buildroot	Version: 2024.02	Provided in the resource package
Qt5	Version: 5.15.11	Provided in the resource package
Cross Compiler Toolchain	aarch64-buildroot-linux-gnu-	Provided in the resource package
System Burning Method	PC-side burning	User tutorial provided
MIPI LCD Driver	MIPI LCD driver	Provided in the resource package

Touch	GT911 capacitive touch screen (only for screens sold by ALIENTEK)	Provided in the resource package
Network	Gigabit Ethernet PHY: YT8531C	Provided in the resource package
USB HOST	1× USB HOST 2.0 1× USB HOST 3.0	Provided in the resource package
RESET Button	Reset function	Provided in the resource package
TF Card	SDMMC driver	Provided in the resource package
LED	GPIO	Provided in the resource package
Audio	ES8390	Provided in the resource package
USB WIFI&BT	RTL8733	Provided in the resource package
Serial Port	USB debug serial port	Provided in the resource package
ADC	ADC driver	Provided in the resource package
PWM	LCD PWM backlight	Provided in the resource package
CAN	2× CAN channels	Provided in the resource package

Table 3.2.1 Factory Linux System Software Resources of the Development Board

This covers the software resources of the ALIENTEK ATK-DLRV1126B development board. We will continue to update the software resources on an ongoing basis.

Chapter 4. Development documents

The ALIENTEK provided the development board ATK-DLRV1126B with a wealth of development documents and system source code software resources. All software resources are available for free download through Baidu Netdisk.

4.1 Download of Materials

Development Board Materials Download Center:
http://www.openedv.com/docs/boards/arm-linux/index.html

Chapter 5. Precautions and maintenance

Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

Chapter 6. After sales service

6.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

6.2 After-sales Support

Technical support: Contact the customer service to obtain the group number.

QQ group: ALIENTEK Rockchip Communication group: 775783660
ALIENTEK RV1126B Technical Support Group:
571769456 (order number required)

Taobao shop: ALIENTEK flagship store

Forum: http://www.openedv.com/forum.php